

Employee360 HRMS

Product Requirements Document (PRD)

Enterprise Human Resource Management System
For Nigerian Organizations

Document Version: 1.0

Date: 25 July 2026

Classification: Internal — Confidential

Table of Contents

1. 1. Document Control
2. 2. Executive Summary
3. 3. Problem Statement & Current-State Pain Points
4. 4. Goals, Objectives & Success Metrics
5. 5. Scope
6. 6. User Personas & Roles
7. 7. User Stories & Use Cases
8. 8. Functional Requirements
9. 9. Non-Functional Requirements
10. 10. Workflow & Approval Logic
11. 11. Data Model Overview
12. 12. Integration Requirements
13. 13. Security & Compliance
14. 14. Reporting & Analytics Requirements
15. 15. Technical Architecture Overview
16. 16. Assumptions, Constraints & Dependencies
17. 17. Release Plan / Phased Roadmap
18. 18. Risks & Mitigations
19. 19. Glossary of Terms
20. 20. Appendix

1. Document Control

Field	Value
Document Title	Employee360 HRMS — Product Requirements Document
Document ID	PRD-E360-001
Version	1.0
Status	Draft — For Review
Author	Product Management / Solutions Architecture
Creation Date	25 July 2026
Last Updated	25 July 2026
Product Name	Employee360 HRMS
Target Market	Nigerian Enterprises (50–5,000+ employees)
Distribution	Internal — Engineering, QA, HR, IT, Executive Stakeholders

Revision History

Version	Date	Author	Description
0.1	2026-07-01	Product Management	Initial outline and stakeholder interviews
0.5	2026-07-15	Solutions Architecture	Technical architecture and data model draft
1.0	25 Jul 2026	Product Management	Full PRD for Phase 1–3 delivery

Approvers

Name	Role	Signature	Date	Status
[TBD]	Head of HR			Pending
[TBD]	Chief Information Officer			Pending
[TBD]	Chief Financial Officer			Pending
[TBD]	Engineering Lead			Pending
[TBD]	Data Protection Officer			Pending

2. Executive Summary

Employee360 HRMS is a custom-built, enterprise-grade Human Resource Management System designed specifically for organizations operating in Nigeria. The platform replaces fragmented spreadsheets, paper-based workflows, and disconnected point solutions with a unified digital system covering the full employee lifecycle — from hire to retire.

The system will be delivered in three phased releases. Phase 1 (MVP) establishes the employee master data foundation, role-based security, leave management, and employee/manager self-service portals. Phase 2 adds attendance and time management, Nigeria-compliant payroll

(PAYE, Pension, NHF, NSITF), and operational analytics. Phase 3 extends the platform with recruitment, onboarding, and performance management capabilities.

Built on Microsoft technologies — ASP.NET Core, Angular, SQL Server, and Azure — Employee360 aligns with enterprise IT standards, supports future Azure AD/SSO integration, and is architected for scalability, auditability, and regulatory compliance under the Nigeria Data Protection Regulation (NDPR).

Key Business Outcomes:

- Reduce HR administrative processing time by 60% within 12 months of Phase 1 go-live.
- Achieve 95%+ employee adoption of self-service for leave requests within 6 months.
- Eliminate parallel payroll spreadsheets and reduce payroll processing cycle from days to hours.
- Provide real-time workforce visibility to HR leadership and line managers.
- Establish a compliant, auditable system of record for all employee and payroll data.

3. Problem Statement & Current-State Pain Points

Nigerian enterprises — particularly those with 50+ employees across multiple locations — face significant operational friction in managing human resources. Current-state processes are predominantly manual, decentralized, and error-prone, creating compliance risk, employee dissatisfaction, and inability to make data-driven workforce decisions.

Current-State Pain Points

Pain Point	Description	Business Impact
Fragmented employee data	Employee records maintained across Excel, email, and paper files with no single source of truth.	Data inconsistency; inability to produce accurate headcount reports; onboarding delays.
Manual leave management	Leave requests via email/WhatsApp; balances tracked in spreadsheets; no audit trail.	Policy violations; payroll errors from unrecorded leave; manager bottlenecks.
Payroll complexity	Nigerian statutory deductions (PAYE, Pension 8%/10%, NHF 2.5%, NSITF) calculated manually.	Compliance risk with FIRS/LASRRRA; calculation errors; delayed salary payments.
No self-service	Employees depend on HR for profile updates, leave balances, payslips.	HR team overwhelmed; poor employee experience; low transparency.
Attendance gaps	No centralized time tracking; biometric devices not integrated.	Inaccurate payroll inputs; inability to enforce shift policies.
Recruitment disconnection	Hiring tracked separately from employee records.	Duplicate data entry; poor candidate experience; slow time-to-hire.
Limited reporting	No real-time dashboards; reports require manual compilation.	Delayed decisions; inability to track attrition, leave trends, or payroll costs.
Compliance exposure	Inadequate audit trails; PII stored insecurely; NDPR obligations unmet.	Regulatory penalties; reputational risk; data breach exposure.
Scalability constraints	Current tools do not scale with	Process breakdown as organization

	headcount growth or multi-location operations.	grows; increased HR headcount needed.
--	--	---------------------------------------

4. Goals, Objectives & Success Metrics

Strategic Goals

Goal ID	Goal	Description
G-01	Unified HR Platform	Establish Employee360 as the single system of record for all employee and HR transactional data.
G-02	Regulatory Compliance	Ensure payroll and data handling comply with Nigerian tax, pension, and data protection regulations.
G-03	Employee Empowerment	Enable employees and managers to complete routine HR tasks without HR intervention.
G-04	Operational Efficiency	Automate approval workflows, calculations, and notifications to reduce manual processing.
G-05	Workforce Intelligence	Provide leadership with accurate, timely analytics on headcount, attrition, leave, and payroll costs.

Objectives & Measurable KPIs

Objective	KPI	Baseline	Target	Measurement Method	Timeline
Reduce HR admin burden	Avg. time to process leave request	3–5 days	< 24 hours	System timestamp: submit → final approval	6 months post Phase 1
Employee adoption	% leave requests via self-service	< 20%	≥ 95%	System transaction logs	6 months post Phase 1
Data accuracy	Employee record error rate	Unknown (~15% est.)	< 2%	Quarterly data audit sample	12 months post Phase 1
Payroll accuracy	Payroll adjustment rate per cycle	> 10%	< 2%	Payroll correction count / total payslips	3 months post Phase 2
Payroll cycle time	Days from cut-off to bank file generation	5–7 days	≤ 2 days	Payroll run timestamps	3 months post Phase 2
Attendance compliance	% attendance records captured digitally	< 30%	≥ 90%	Attendance records / expected records	6 months post Phase 2
Manager engagement	% approvals completed within SLA	N/A	≥ 85%	Approval workflow timestamps	6 months post Phase 1
System availability	Uptime during business hours	N/A	≥ 99.5%	Azure Monitor / Application	Ongoing

				Insights	
User satisfaction	Net Promoter Score (employees)	N/A	≥ 40	Quarterly pulse survey	12 months post Phase 1
Time-to-hire	Days from job posting to offer acceptance	45+ days	≤ 30 days	Recruitment module timestamps	6 months post Phase 3

5. Scope

5.1 In-Scope — Phase 1 (MVP)

- Employee Management: core master data, profiles, documents, org chart, departments, positions
- Authentication, Roles & Permissions (RBAC) with JWT
- Leave Management: configurable leave types, approval workflows, balances and accrual
- Employee Self-Service (ESS) and Manager Self-Service (MSS) dashboards
- Email notifications (SMTP) for key workflow events
- Audit logging for sensitive data changes
- Basic admin configuration (company profile, working calendar, public holidays)

5.2 In-Scope — Phase 2

- Attendance & Time Management: clock-in/out, shifts, timesheets, device integration readiness
- Nigeria-specific Payroll: PAYE, Pension (8% employee / 10% employer), NHF, NSITF, salary structures, deductions, payslips, bank payment files
- Reports & Analytics: headcount, attrition, leave trends, payroll cost dashboards
- Payroll approval workflow and payroll period locking

5.3 In-Scope — Phase 3

- Recruitment & Onboarding: job postings, applicant tracking, offer letters, onboarding checklists
- Performance Management: goals/KPIs, appraisals, 360-degree reviews

5.4 Out-of-Scope (All Phases / Later Roadmap)

Item	Rationale	Planned Phase
Training & Development (LMS)	Requires separate content management; lower priority than core HR.	Later
Benefits & Compliance module	Complex benefits administration deferred until payroll stable.	Later
Native mobile applications (iOS/Android)	Responsive web meets MVP; native apps evaluated post-adoption.	Later
Full ERP/Accounting integration	Bank payment file export provided;	Later

	full GL integration is future.	
Multi-country payroll	Nigeria-only in initial releases.	Later
Biometric hardware procurement	Integration APIs provided; hardware supplied by client.	Phase 2 (integration only)
Azure AD / SSO implementation	Architecture supports; implementation deferred.	Post Phase 1
Employee wellness / engagement surveys	Not core HRMS scope.	Later
Union/Collective bargaining management	Client-specific; requires separate analysis.	TBD

6. User Personas & Roles

Persona	Description	Primary Goals	Key Permissions
HR Admin	Day-to-day HR operations staff managing employee records, leave policies, and system configuration.	Maintain accurate employee data; configure leave policies; support employees; generate reports.	Full CRUD on employee records; configure leave types/policies; view all employee data; manage documents; run HR reports.
HR Manager	Head of HR / HR Business Partner with oversight of HR operations and policy.	Ensure policy compliance; approve escalated requests; monitor workforce metrics.	All HR Admin permissions plus: approve escalated leave; view analytics dashboards; approve profile changes; lock payroll periods (Phase 2).
Line Manager	Department head or team lead with direct reports.	Manage team leave approvals; view team attendance; conduct performance reviews (Phase 3).	View direct/indirect reports; approve/reject leave; view team calendar; view team attendance (Phase 2); complete appraisals (Phase 3).
Employee	All staff members using self-service.	View/update personal info; request leave; access payslips; view attendance.	View own profile; submit leave requests; view leave balance; download payslips (Phase 2); clock in/out (Phase 2); view own documents.
Payroll Officer	Finance/HR staff responsible for salary processing.	Run accurate, compliant payroll; generate bank files and payslips.	Manage salary structures; run payroll; generate payslips; export bank payment files; view payroll reports. No access to recruitment (unless dual-role).
System/IT Admin	Technical administrator managing users, roles, system settings.	Maintain system security; manage user accounts; configure integrations.	Manage roles/permissions; system configuration; view audit logs; manage integrations; no access to employee PII unless granted.
Executive / Viewer	C-suite or senior leadership (read-only).	Monitor workforce KPIs and trends.	View executive dashboards and reports only; no transactional

			access.
--	--	--	---------

Role Hierarchy & Inheritance

Roles are flat assignments (a user may hold multiple roles). Permissions are additive. Manager permissions are scoped to direct and indirect reports via the reporting hierarchy. Department-scoped data access may be configured for HR Business Partners in a future enhancement.

7. User Stories & Use Cases

7.1 Authentication & RBAC

As a Employee, I want to log in securely with my corporate email and password, so that I can access my self-service portal.

Acceptance Criteria:

- Given I am a registered active employee, When I enter valid credentials, Then I am authenticated and redirected to my ESS dashboard.
- Given I enter invalid credentials 5 times, When I attempt a 6th login, Then my account is temporarily locked for 15 minutes.
- Given my account is deactivated, When I attempt to log in, Then I receive an error and cannot access the system.
- Given I am authenticated, When my JWT expires, Then I am prompted to re-authenticate or use refresh token.

As a System Admin, I want to assign roles and permissions to users, so that access is controlled according to job function.

Acceptance Criteria:

- Given I am a System Admin, When I assign the 'Line Manager' role to a user, Then that user can view and approve leave for their direct reports only.
- Given I remove a role from a user, When they next authenticate, Then their permissions reflect the updated role assignment.
- Given a permission change is made, When the change is saved, Then an audit log entry is created with actor, timestamp, and change details.

7.2 Employee Management

As a HR Admin, I want to create and maintain employee master records, so that the organization has a single source of truth for employee data.

Acceptance Criteria:

- Given I am an HR Admin, When I create a new employee with required fields, Then the employee record is saved with status 'Active' and a unique Employee ID is generated.
- Given an employee exists, When I update their department or manager, Then the org chart and reporting relationships are updated immediately.
- Given I deactivate an employee, When I set status to 'Terminated', Then the employee cannot log in but their historical records are retained.
- Given I upload a document to an employee profile, When the upload completes, Then the document is stored securely and visible per RBAC rules.

As a Employee, I want to view my profile and employment details, so that I can verify my information is correct.

Acceptance Criteria:

- Given I am logged in, When I navigate to My Profile, Then I see my personal details, employment info, manager, department, and job title.
- Given I request a profile change (e.g., phone number), When I submit the request, Then it is routed to HR for approval and I receive a notification on decision.

As a Line Manager, I want to view my team's org structure and profiles, so that I can understand my reporting structure and team composition.

Acceptance Criteria:

- Given I am a Line Manager, When I open the Team view, Then I see all direct reports with name, title, department, and status.
- Given I view the Org Chart, When I click on a team member, Then I see their basic profile (excluding sensitive fields like bank details).

7.3 Leave Management

As a Employee, I want to apply for leave and view my balance, so that I can plan time off without emailing HR.

Acceptance Criteria:

- Given I have 10 days annual leave balance, When I apply for 3 days leave, Then my pending balance shows 7 days upon submission.
- Given my leave request overlaps a public holiday, When I submit the request, Then public holidays are excluded from working days calculation.
- Given my manager approves my leave, When I view my requests, Then status shows 'Approved' and I receive an email notification.
- Given my leave exceeds available balance, When I submit the request, Then the system prevents submission and displays an error.

As a Line Manager, I want to approve or reject leave requests from my team, so that team coverage is managed efficiently.

Acceptance Criteria:

- Given a direct report submits leave, When I open my approval queue, Then I see the request with dates, type, balance, and reason.
- Given I approve a request, When I confirm, Then the employee is notified and the leave is reflected on the team calendar.
- Given I do not act within 48 hours, When the SLA expires, Then the request is escalated to the next-level manager or HR.

7.4 Attendance & Time (Phase 2)

As a Employee, I want to clock in and clock out, so that my attendance is recorded accurately for payroll.

Acceptance Criteria:

- Given I am within an allowed location/shift window, When I clock in, Then an attendance record is created with timestamp and source (web/device).
- Given I forgot to clock out, When I notify my manager, Then the manager or HR can submit a correction with reason (audit logged).

As a HR Admin, I want to configure shifts and work schedules, so that attendance rules align with company policy.

Acceptance Criteria:

- Given I define a shift (08:00–17:00, Mon–Fri), When assigned to a department, Then employees in that department are evaluated against that schedule.
- Given an employee clocks in 30 minutes late, When the daily record is processed, Then status reflects 'Late' per configured grace period.

7.5 Payroll (Phase 2)

As a Payroll Officer, I want to run monthly payroll with Nigerian statutory deductions, so that employees are paid accurately and compliantly.

Acceptance Criteria:

- Given an approved payroll period, When I initiate a payroll run, Then PAYE, Pension (8%/10%), NHF (2.5%), and NSITF are calculated per configured rules.
- Given payroll is calculated, When I review the summary, Then I can see gross pay, each deduction line item, and net pay per employee.
- Given payroll is approved, When I generate payslips, Then each employee can download their PDF payslip from self-service.

- Given payroll is finalized, When I export the bank payment file, Then it is formatted per configured bank template (e.g., NIBSS/NEFT CSV).

As a Employee, I want to view and download my payslips, so that I have transparent access to my compensation details.

Acceptance Criteria:

- Given payroll has been finalized for a period, When I navigate to My Payslips, Then I see a list of available payslips by month.
- Given I select a payslip, When I download it, Then I receive a PDF showing earnings, statutory deductions, and net pay in NGN (₦).

7.6 Recruitment & Onboarding (Phase 3)

As a HR Admin, I want to post job openings and track applicants, so that hiring is managed within the same platform as employee records.

Acceptance Criteria:

- Given I create a job posting, When I publish it, Then it is visible on the internal/external careers page.
- Given a candidate applies, When I update their stage to 'Interview', Then the hiring team is notified.
- Given a candidate is marked 'Hired', When I initiate onboarding, Then a new employee record is pre-populated from candidate data.

7.7 Performance Management (Phase 3)

As a Line Manager, I want to set goals and conduct appraisals for my team, so that performance is tracked and documented systematically.

Acceptance Criteria:

- Given a review cycle is active, When I set KPIs for a direct report, Then the employee can view their goals in self-service.
- Given the employee completes self-assessment, When I submit my manager review, Then the final rating is calculated per configured weighting.
- Given a 360 review is configured, When peer feedback is submitted, Then it is aggregated anonymously in the review summary.

7.8 Employee & Manager Self-Service (ESS/MSS)

As a Employee, I want to see a personalized dashboard when I log in, so that I have immediate visibility of my HR tasks and status.

Acceptance Criteria:

- Given I log in as an Employee, When the ESS dashboard loads, Then I see widgets for leave balance, pending requests, notifications, and quick actions.
- Given I have no pending items, When I view the dashboard, Then I see an empty state with guidance to apply for leave or update my profile.

As a Line Manager, I want to see all pending approvals in one place, so that I can action team requests efficiently without searching emails.

Acceptance Criteria:

- Given I have 3 pending leave approvals, When I open MSS Approvals, Then I see all 3 with employee name, dates, type, and days requested.
- Given I bulk-approve eligible requests, When I select multiple and confirm, Then each is processed individually with audit entries.

7.9 Administration & Configuration

As a HR Admin, I want to configure company public holidays, so that leave calculations exclude non-working days correctly.

Acceptance Criteria:

- Given I add a public holiday for 1 October (Independence Day), When an employee applies leave spanning that date, Then 1 October is excluded from working days.
- Given I import holidays for the full year, When saved, Then they appear on the leave calendar for all users.

As a System Admin, I want to view system audit logs, so that security incidents and data changes can be investigated.

Acceptance Criteria:

- Given I filter audit logs by employee entity and date range, When results load, Then I see all create/update/delete actions with actor, timestamp, and field-level changes.
- Given I export audit logs, When I click Export, Then I receive a CSV file for the filtered period.

7.10 Use Cases (Narrative)

UC-001: Onboard New Employee

Primary Actor: HR Admin. Precondition: Departments and positions exist. Flow: HR Admin creates employee record with personal and employment details → System generates Employee ID → System creates user account and sends welcome email with password setup link → Employee logs in and completes profile → HR uploads onboarding documents. Postcondition: Employee appears in directory with Active status.

UC-002: Monthly Payroll Processing

Primary Actor: Payroll Officer. Precondition: Attendance and leave data finalized for period; salary structures assigned. Flow: Payroll Officer initiates payroll run → System calculates gross, PAYE, Pension, NHF, NSITF, and custom deductions → Payroll Officer reviews exceptions → Submits for HR Manager approval → Finance approves → System generates payslips and bank file → Period locked. Postcondition: Employees can access payslips; bank file ready for upload.

UC-003: Employee Resignation & Offboarding

Primary Actor: HR Admin. Flow: HR Admin sets employee status to Resigned with last working day → System revokes login access on last working day → Leave balance finalized → Offboarding checklist initiated (Phase 3) → Employee record retained for audit. Postcondition: Employee excluded from active payroll and headcount.

8. Functional Requirements

Priority Legend: Must = mandatory for release; Should = important but can defer within phase; Could = desirable; Won't = explicitly excluded this phase.

FR ID	Module	Requirement	Priority	Phase
FR-AUTH-001	Authentication	System shall support email + password login with JWT access tokens and refresh tokens.	Must	Phase 1
FR-AUTH-002	Authentication	System shall enforce password policy: min 8 chars, uppercase, lowercase, number, special character.	Must	Phase 1
FR-AUTH-003	Authentication	System shall lock accounts after 5 failed login attempts for 15 minutes.	Must	Phase 1
FR-AUTH-004	Authentication	System shall support password reset via email token (expires in 1 hour).	Must	Phase 1
FR-AUTH-005	Authentication	System architecture shall support future Azure AD/OIDC SSO integration without schema changes.	Must	Phase 1
FR-AUTH-006	RBAC	System shall support role-based permissions configurable per module and action (CRUD).	Must	Phase 1
FR-AUTH-007	RBAC	System shall support multi-role assignment per user with additive permissions.	Must	Phase 1

FR-AUTH-008	RBAC	Manager-scoped data access shall be enforced based on reporting hierarchy.	Must	Phase 1
FR-EMP-001	Employee Mgmt	System shall auto-generate unique Employee ID on creation (configurable format, e.g., EMP-00001).	Must	Phase 1
FR-EMP-002	Employee Mgmt	System shall store core personal data: name, DOB, gender, nationality, marital status, contact details, address, NIN (optional).	Must	Phase 1
FR-EMP-003	Employee Mgmt	System shall store employment data: department, position, grade, manager, join date, employment type, work location.	Must	Phase 1
FR-EMP-004	Employee Mgmt	System shall support employee status lifecycle: Draft, Active, On Leave, Suspended, Resigned, Terminated.	Must	Phase 1
FR-EMP-005	Employee Mgmt	System shall soft-delete employee records (IsDeleted flag); no hard delete of employee data.	Must	Phase 1
FR-EMP-006	Employee Mgmt	System shall maintain audit columns: CreatedBy, CreatedAt, ModifiedBy, ModifiedAt on all entities.	Must	Phase 1
FR-EMP-007	Employee Mgmt	System shall support employee document upload (PDF, JPG, PNG; max 10MB) with category tagging.	Must	Phase 1
FR-EMP-008	Employee Mgmt	System shall display interactive org chart based on reporting hierarchy.	Should	Phase 1
FR-EMP-009	Employee Mgmt	System shall support department CRUD with parent-child hierarchy.	Must	Phase 1
FR-EMP-010	Employee Mgmt	System shall support position/job title CRUD linked to	Must	Phase 1

		departments and grades.		
FR-EMP-011	Employee Mgmt	System shall support employee profile change requests with HR approval workflow.	Should	Phase 1
FR-EMP-012	Employee Mgmt	System shall support bulk employee import via CSV template.	Could	Phase 1
FR-EMP-013	Employee Mgmt	System shall store emergency contact and next-of-kin details.	Must	Phase 1
FR-EMP-014	Employee Mgmt	System shall store bank account details (bank name, account number, account name) for payroll.	Must	Phase 1
FR-LV-001	Leave Mgmt	System shall support configurable leave types (Annual, Sick, Maternity, Paternity, Compassionate, Unpaid).	Must	Phase 1
FR-LV-002	Leave Mgmt	System shall support leave policy configuration: annual entitlement, accrual frequency, carry-forward rules, max carry-forward, probation exclusion.	Must	Phase 1
FR-LV-003	Leave Mgmt	System shall calculate working days excluding weekends and configured public holidays.	Must	Phase 1
FR-LV-004	Leave Mgmt	System shall track leave balance per employee per leave type per year.	Must	Phase 1
FR-LV-005	Leave Mgmt	System shall support leave application with start date, end date, type, reason, and attachment (optional).	Must	Phase 1
FR-LV-006	Leave Mgmt	System shall enforce leave balance validation before submission.	Must	Phase 1
FR-LV-007	Leave Mgmt	System shall route leave requests through configurable approval chain: Employee → Line	Must	Phase 1

		Manager → HR (optional).		
FR-LV-008	Leave Mgmt	System shall support approve, reject, and return-for-revision actions with comments.	Must	Phase 1
FR-LV-009	Leave Mgmt	System shall auto-escalate pending approvals after configurable SLA (default 48 hours).	Should	Phase 1
FR-LV-010	Leave Mgmt	System shall display team leave calendar for managers.	Must	Phase 1
FR-LV-011	Leave Mgmt	System shall send email notifications on leave submit, approve, reject, and escalate.	Must	Phase 1
FR-LV-012	Leave Mgmt	HR Admin shall be able to manually adjust leave balances with reason (audit logged).	Must	Phase 1
FR-ESS-001	Self-Service	ESS dashboard shall show leave balance, pending requests, and recent notifications.	Must	Phase 1
FR-ESS-002	Self-Service	ESS shall provide quick actions: Apply Leave, View Profile, View Documents.	Must	Phase 1
FR-MSS-001	Manager Self-Service	MSS dashboard shall show pending approvals, team on leave today, and team headcount.	Must	Phase 1
FR-MSS-002	Manager Self-Service	MSS shall provide team list with search and filter by status.	Must	Phase 1
FR-ATT-001	Attendance	System shall support web-based clock-in/clock-out with timestamp and IP/device logging.	Must	Phase 2
FR-ATT-002	Attendance	System shall support shift definitions (start, end, grace period, break duration).	Must	Phase 2
FR-ATT-003	Attendance	System shall calculate daily attendance status: Present, Absent, Late, Half-Day, On Leave, Holiday.	Must	Phase 2

FR-ATT-004	Attendance	System shall support timesheet entry and approval for non-shift employees.	Should	Phase 2
FR-ATT-005	Attendance	System shall provide REST API endpoint for biometric device integration (clock event ingestion).	Must	Phase 2
FR-ATT-006	Attendance	System shall support manual attendance correction by HR/Manager with mandatory reason.	Must	Phase 2
FR-ATT-007	Attendance	System shall support overtime calculation per configured rules.	Should	Phase 2
FR-ATT-008	Attendance	System shall integrate approved leave into attendance records automatically.	Must	Phase 2
FR-PAY-001	Payroll	System shall support salary structure definition: Basic, Housing, Transport, and other allowances.	Must	Phase 2
FR-PAY-002	Payroll	System shall support employee salary assignment with effective date and history.	Must	Phase 2
FR-PAY-003	Payroll	System shall calculate PAYE tax per Nigeria Tax Act bands (configurable annually).	Must	Phase 2
FR-PAY-004	Payroll	System shall calculate Pension: 8% employee contribution + 10% employer contribution on qualifying emoluments.	Must	Phase 2
FR-PAY-005	Payroll	System shall calculate NHF at 2.5% of basic salary (employee contribution).	Must	Phase 2
FR-PAY-006	Payroll	System shall calculate NSITF employer contribution per configured rate on total emoluments.	Must	Phase 2
FR-PAY-007	Payroll	System shall support additional deductions: loans,	Must	Phase 2

		cooperative, union dues, custom deductions.		
FR-PAY-008	Payroll	System shall support payroll run workflow: Draft → Calculated → Reviewed → Approved → Finalized.	Must	Phase 2
FR-PAY-009	Payroll	System shall generate itemized payslip PDF per employee per period.	Must	Phase 2
FR-PAY-010	Payroll	System shall generate bank payment file (CSV/TXT) for salary disbursement.	Must	Phase 2
FR-PAY-011	Payroll	System shall lock finalized payroll periods against modification.	Must	Phase 2
FR-PAY-012	Payroll	System shall factor approved unpaid leave and attendance into payroll calculations.	Must	Phase 2
FR-PAY-013	Payroll	System shall store Pension PIN (PFA details) per employee.	Must	Phase 2
FR-PAY-014	Payroll	System shall produce statutory remittance reports (PAYE, Pension, NHF summaries).	Should	Phase 2
FR-RPT-001	Reporting	System shall provide headcount report by department, location, employment type, gender.	Must	Phase 2
FR-RPT-002	Reporting	System shall provide attrition report with voluntary/involuntary breakdown.	Must	Phase 2
FR-RPT-003	Reporting	System shall provide leave utilization and balance report.	Must	Phase 2
FR-RPT-004	Reporting	System shall provide payroll cost summary by department and period.	Must	Phase 2
FR-RPT-005	Reporting	All reports shall support date range filter and export to CSV/Excel.	Must	Phase 2
FR-RPT-006	Reporting	System shall provide executive dashboard with KPI widgets.	Should	Phase 2

FR-REC-001	Recruitment	System shall support job posting CRUD with title, description, department, requirements, closing date.	Must	Phase 3
FR-REC-002	Recruitment	System shall support applicant pipeline stages: Applied, Screening, Interview, Offer, Hired, Rejected.	Must	Phase 3
FR-REC-003	Recruitment	System shall support resume/CV upload per candidate.	Must	Phase 3
FR-REC-004	Recruitment	System shall generate offer letter from configurable template.	Should	Phase 3
FR-REC-005	Recruitment	System shall convert hired candidate to employee record (pre-filled onboarding).	Must	Phase 3
FR-REC-006	Recruitment	System shall support onboarding checklist with assignable tasks and due dates.	Must	Phase 3
FR-PERF-001	Performance	System shall support review cycle configuration (annual, mid-year, probation).	Must	Phase 3
FR-PERF-002	Performance	System shall support goal/KPI setting per employee per cycle.	Must	Phase 3
FR-PERF-003	Performance	System shall support self-assessment and manager assessment forms.	Must	Phase 3
FR-PERF-004	Performance	System shall calculate weighted final rating from configured criteria.	Must	Phase 3
FR-PERF-005	Performance	System shall support 360-degree feedback with anonymous peer input.	Should	Phase 3
FR-PERF-006	Performance	System shall maintain performance history on employee profile.	Must	Phase 3
FR-ADM-001	Admin	System shall support company profile configuration (name, logo, address, TIN, RC number).	Must	Phase 1
FR-ADM-002	Admin	System shall support	Must	Phase 1

		public holiday calendar management per year.		
FR-ADM-003	Admin	System shall support configurable email templates for workflow notifications.	Should	Phase 1
FR-ADM-004	Admin	System shall provide global audit log viewer with filter by user, entity, date range.	Must	Phase 1

9. Non-Functional Requirements

NFR ID	Category	Requirement	Target/Standard
NFR-PERF-001	Performance	Page load time for dashboard (95th percentile)	≤ 3 seconds on 10 Mbps connection
NFR-PERF-002	Performance	API response time for standard CRUD (95th percentile)	≤ 500 ms
NFR-PERF-003	Performance	Payroll run for 2,000 employees	≤ 10 minutes
NFR-PERF-004	Performance	Concurrent authenticated users	≥ 500 without degradation
NFR-SCAL-001	Scalability	Support headcount growth without architecture change	Up to 10,000 employees
NFR-SCAL-002	Scalability	Database design supports horizontal read scaling (Azure SQL read replicas)	Phase 2 readiness
NFR-SEC-001	Security	All traffic encrypted via TLS 1.2+	Mandatory
NFR-SEC-002	Security	Passwords hashed with bcrypt or ASP.NET Identity PBKDF2	Mandatory
NFR-SEC-003	Security	JWT access token expiry	15 minutes (configurable)
NFR-SEC-004	Security	Sensitive PII fields encrypted at rest (bank account, NIN)	AES-256
NFR-SEC-005	Security	OWASP Top 10 mitigations implemented	Mandatory
NFR-AVAIL-001	Availability	System uptime during business hours (Mon–Sat 07:00–20:00 WAT)	≥ 99.5%
NFR-AVAIL-002	Availability	Planned maintenance window	Sundays 00:00–04:00 WAT with 48hr notice
NFR-AVAIL-003	Disaster Recovery	RPO / RTO	RPO ≤ 1 hour; RTO ≤ 4 hours
NFR-USAB-001	Usability	Responsive design for desktop, tablet, and mobile browsers	Mandatory

NFR-USAB-002	Usability	Microsoft Fluent UI design language	Mandatory
NFR-USAB-003	Usability	New employee completes first leave request without training	≤ 5 minutes (usability test)
NFR-ACC-001	Accessibility	WCAG 2.1 Level AA compliance for all employee-facing screens	Mandatory
NFR-LOC-001	Localization	Currency displayed as Nigerian Naira (₦ / NGN) with comma separators	Mandatory
NFR-LOC-002	Localization	Date format: DD/MM/YYYY; Timezone: WAT (UTC+1)	Mandatory
NFR-LOC-003	Localization	English (en-NG) as primary language; architecture supports future localization	Phase 1
NFR-AUD-001	Auditability	All create/update/delete on sensitive entities logged with actor, timestamp, old/new values	Mandatory
NFR-AUD-002	Auditability	Audit logs retained minimum 7 years	Mandatory
NFR-RET-001	Data Retention	Terminated employee data retained per company policy (default 7 years)	Configurable
NFR-RET-002	Data Retention	Soft-deleted records excluded from UI but retained in database	Mandatory
NFR-MAINT-001	Maintainability	Clean Architecture with vertical slice organization	Mandatory
NFR-MAINT-002	Maintainability	Code coverage for business logic (unit tests)	≥ 80%

10. Workflow & Approval Logic

10.1 Leave Approval Workflow

Flow Diagram (Text):

[Employee] → Submit Leave Request → [System validates balance & dates]

- └ Invalid → Return error to Employee
- └ Valid → Notify [Line Manager] → Pending Approval
 - └ Approve → (If HR approval required?)
 - | └ Yes → Notify [HR Admin] → Approve/Reject
 - | └ No → Deduct balance → Notify Employee → Update Team Calendar
 - └ Reject → Notify Employee with reason

- └─ Return for Revision → Notify Employee
- └─ No action within SLA (48h) → Escalate to [Next-Level Manager or HR]

Step	Actor	Action	System Behavior
1	Employee	Submit leave request	Validate balance, dates, overlaps; set status = Pending; notify manager
2	Line Manager	Review request	View team calendar, balance, history; approve/reject/return
3	HR Admin (if configured)	Final approval	Required for maternity, unpaid, or exceeds threshold days
4	System	Escalation	After 48h pending: notify escalation recipient; log escalation event
5	System	Completion	Update balance, attendance preview, send email notification

10.2 Payroll Approval Workflow

- [Payroll Officer] → Initiate Payroll Run (Draft) → System calculates all employees
 → [Payroll Officer] Reviews exceptions/variance report
 → Submit for Approval → [HR Manager] reviews summary
 → Approve → [CFO/Finance] final approval (configurable)
 → Finalize → Generate payslips + bank file → Lock period
 → Notify employees: payslips available

Status	Description	Allowed Actions	Next Status
Draft	Payroll run initiated, calculations in progress	Calculate, Cancel	Calculated
Calculated	All employee pay computed; exceptions flagged	Review, Recalculate, Submit	Pending Approval
Pending Approval	Awaiting HR Manager / Finance approval	Approve, Reject	Approved / Calculated
Approved	Approved for disbursement	Finalize	Finalized
Finalized	Payslips generated, bank file exported, period locked	None (reversal requires admin override)	—

10.3 Profile Change Request Workflow

- [Employee] → Submit change (phone, address, etc.) → [HR Admin] Review
 └─ Approve → Update employee record → Notify employee
 └─ Reject → Notify employee with reason

10.4 Recruitment-to-Employee Workflow (Phase 3)

[HR Admin] → Create Job Posting → [Candidate] Applies → Pipeline stages
→ Offer extended → Candidate accepts → [HR Admin] Initiate Onboarding
→ Create Employee record (pre-filled) → Onboarding checklist tasks assigned
→ All tasks complete → Employee status = Active

11. Data Model Overview

11.1 Key Entities

Entity	Description	Key Attributes
Company	Organization tenant	Name, RCNumber, TIN, Address, LogoUrl, DefaultCurrency (NGN)
Department	Organizational unit	Name, Code, ParentDepartmentId, HeadEmployeeId, IsActive
Position	Job title/role	Title, Code, DepartmentId, GradeId, Description
Grade	Salary/level grade	Name, Level, MinSalary, MaxSalary
Employee	Core person record	EmployeeId, FirstName, LastName, Email, DOB, Gender, NIN, Status, DepartmentId, PositionId, ManagerId, JoinDate
EmployeeBankAccount	Payment details	EmployeeId, BankName, AccountNumber (encrypted), AccountName
EmployeeDocument	Uploaded files	EmployeeId, Category, FileName, StoragePath, ExpiryDate
User	Auth account	Email, PasswordHash, EmployeeId, IsActive, LastLoginAt
Role	RBAC role	Name, Description
Permission	Granular permission	Module, Action, Resource
RolePermission	Role-permission mapping	RoleId, PermissionId
UserRole	User-role assignment	UserId, RoleId
LeaveType	Leave category	Name, Code, IsPaid, RequiresAttachment, Color
LeavePolicy	Policy rules	LeaveTypeId, AnnualEntitlement, AccrualFrequency, CarryForwardMax, ProbationMonths
LeaveBalance	Employee balance	EmployeeId, LeaveTypeId, Year, Entitled, Used, Pending, CarriedForward
LeaveRequest	Leave application	EmployeeId, LeaveTypeId, StartDate, EndDate, Days, Reason, Status, ApproverId
LeaveApproval	Approval step record	LeaveRequestId, ApproverId, Action, Comments, ActionDate
PublicHoliday	Holiday calendar	Date, Name, Year
AttendanceRecord	Daily attendance	EmployeeId, Date, ClockIn, ClockOut, Status, Source, ShiftId
Shift	Work shift definition	Name, StartTime, EndTime, GraceMinutes, BreakMinutes
Timesheet	Time entry	EmployeeId, WeekStart, TotalHours, Status

SalaryStructure	Compensation template	Name, Basic, Housing, Transport, OtherAllowances
EmployeeSalary	Employee pay assignment	EmployeeId, SalaryStructureId, GrossSalary, EffectiveDate
PayrollRun	Payroll batch	PeriodStart, PeriodEnd, Status, RunBy, ApprovedBy, FinalizedAt
Payslip	Employee pay record	PayrollRunId, EmployeeId, GrossPay, PAYE, Pension, NHF, NSITF, NetPay, PdfPath
PayrollDeduction	Custom deduction	EmployeeId, Type, Amount, IsRecurring
JobPosting	Open position	Title, DepartmentId, Description, Status, ClosingDate
Candidate	Applicant	JobPostingId, Name, Email, Phone, Stage, ResumePath
OnboardingTask	Onboarding checklist item	EmployeeId, TaskName, AssignedTo, DueDate, IsCompleted
ReviewCycle	Performance period	Name, StartDate, EndDate, Status
EmployeeGoal	KPI/goal	EmployeeId, ReviewCycleId, Title, Weight, TargetValue, ActualValue
PerformanceReview	Appraisal record	EmployeeId, ReviewCycleId, SelfRating, ManagerRating, FinalRating, Status
AuditLog	Change audit trail	EntityName, EntityId, Action, OldValues, NewValues, UserId, Timestamp
Notification	In-app notification	UserId, Title, Message, IsRead, CreatedAt

11.2 Entity Relationships (ER Description)

- Company has many Departments; Department has optional parent Department (self-referencing hierarchy).
- Department has many Positions; Position belongs to one Department and optional Grade.
- Employee belongs to one Department, one Position, and one Manager (self-referencing Employee).
- Employee has one User account (1:1) for authentication.
- Employee has many EmployeeDocuments, LeaveBalances, LeaveRequests, AttendanceRecords, Payslips.
- LeaveType has one LeavePolicy; LeaveBalance is unique per Employee + LeaveType + Year.
- LeaveRequest has many LeaveApprovals (one per approval step).
- SalaryStructure is assigned to Employees via EmployeeSalary (with effective date history).
- PayrollRun has many Payslips; each Payslip belongs to one Employee.
- JobPosting has many Candidates; hired Candidate converts to Employee.
- ReviewCycle has many EmployeeGoals and PerformanceReviews.
- All entities include audit columns (CreatedBy, CreatedAt, ModifiedBy, ModifiedAt) and IsDeleted for soft delete.

12. Integration Requirements

Integration	Purpose	Protocol/Method	Phase	Notes
SMTP Email (SendGrid / Azure Communication Services)	Workflow notifications, password reset, payslip alerts	SMTP / REST API	Phase 1	HTML email templates; configurable sender address
Biometric Attendance Devices	Ingest clock-in/out events	REST API (webhook/poll)	Phase 2	Vendor-agnostic endpoint; support ZKTeco and similar via middleware
Bank Payment Gateway / File Export	Salary disbursement	CSV/TXT file (NIBSS/NEFT format)	Phase 2	Configurable per bank template; no direct API in initial release
Azure Blob Storage	Document and payslip PDF storage	Azure SDK	Phase 1	SAS token access; encrypted at rest
Azure AD / OIDC (Future)	Enterprise SSO	OpenID Connect	Post Phase 1	Architecture reserves ExternalAuthProvider table
FIRS / State IRS (Future)	Tax filing automation	TBD	Later	Manual export initially; API when available
PFA Pension Platforms (Future)	Pension remittance	TBD	Later	Export pension schedule CSV initially
Microsoft Excel/CSV Export	Report data export	File download	Phase 2	All reports exportable
Hangfire Scheduler	Background jobs: accruals, escalations, payroll	In-process	Phase 1	Recurring jobs for leave accrual and approval escalation

13. Security & Compliance

13.1 Nigeria Data Protection Regulation (NDPR) Compliance

- Lawful basis for processing: employment contract and legitimate business interest.
- Data minimization: collect only HR-necessary personal data.
- Consent captured where required (e.g., optional fields, 360 feedback participation).
- Right to access: employees can view their own data via ESS.
- Right to rectification: profile change request workflow.
- Data breach notification procedure documented (notify NITDA within 72 hours).
- Data Protection Officer (DPO) contact published in system footer.
- Cross-border transfer: all data hosted in Azure region with NDPR-compliant DPA.
- Privacy policy and data processing agreement templates provided.

13.2 Encryption & Data Protection

Data Element	Protection Method
Passwords	Hashed (PBKDF2/bcrypt); never stored in plain text
JWT Tokens	Signed (HMAC-SHA256); short-lived access tokens
Bank account numbers	AES-256 encryption at rest; masked in UI (show last 4

	digits)
NIN	AES-256 encryption at rest; restricted to HR Admin role
Documents (ID, contracts)	Azure Blob Storage with encryption at rest; RBAC-controlled SAS URLs
Data in transit	TLS 1.2+ on all endpoints
SQL Server	Transparent Data Encryption (TDE) enabled on Azure SQL

13.3 RBAC Matrix (Summary)

Full matrix in Appendix A. Key principles:

- Least privilege: users receive minimum permissions for their role.
- Separation of duties: payroll finalize requires different user than payroll calculate.
- Manager scope: automatically limited to direct/indirect reports.
- Audit: all permission changes logged.

14. Reporting & Analytics Requirements

Report ID	Report Name	Audience	Phase	Key Fields
RPT-001	Headcount Summary	HR, Executive	Phase 2	Department, location, gender, employment type, count
RPT-002	Headcount Trend	Executive	Phase 2	Month-over-month headcount, new hires, exits
RPT-003	Attrition Report	HR, Executive	Phase 2	Voluntary/involuntary exits, attrition rate, department breakdown
RPT-004	Leave Balance Report	HR	Phase 1	Employee, leave type, entitled, used, remaining
RPT-005	Leave Utilization	HR, Managers	Phase 2	Department, leave type, days taken, % utilization
RPT-006	Leave History	HR	Phase 1	Employee, dates, type, status, approver
RPT-007	Attendance Summary	HR, Managers	Phase 2	Employee, days present, absent, late, on leave
RPT-008	Payroll Register	Payroll, Finance	Phase 2	Employee, gross, each deduction, net pay
RPT-009	Payroll Cost by Department	Finance, Executive	Phase 2	Department, total gross, total net, employer pension
RPT-010	PAYE Remittance Schedule	Payroll, Finance	Phase 2	Employee, taxable income, PAYE amount
RPT-011	Pension Remittance Schedule	Payroll, Finance	Phase 2	Employee, PFA, PIN, employee/employer contributions
RPT-012	NHF Schedule	Payroll, Finance	Phase 2	Employee, basic

				salary, NHF amount
RPT-013	Bank Payment File	Payroll, Finance	Phase 2	Account number, bank code, net pay, narration
RPT-014	Employee Master List	HR	Phase 1	All core employee fields, exportable CSV
RPT-015	Birthday/Anniversary	HR	Phase 1	Employee, DOB, join date, upcoming dates
RPT-016	Recruitment Pipeline	HR	Phase 3	Job, stage, candidate count, time-in-stage
RPT-017	Performance Rating Distribution	HR, Executive	Phase 3	Department, rating scale, count, average
RPT-018	Onboarding Status	HR	Phase 3	Employee, tasks completed, overdue tasks

Dashboard Widgets

Dashboard	Widgets
ESS Dashboard	Leave balance, pending requests, notifications, quick actions
MSS Dashboard	Pending approvals, team on leave today, team headcount, upcoming birthdays
HR Admin Dashboard	Total headcount, new joiners, exits, pending approvals, leave trends chart
Executive Dashboard	Headcount trend, attrition rate, payroll cost, department breakdown, gender diversity
Payroll Dashboard	Current period status, exceptions, total payroll cost, comparison to prior month

15. Technical Architecture Overview

15.1 Architecture Layers

Layer	Technology	Responsibility
Presentation	Angular 17+ (standalone components, lazy-loaded feature modules)	UI, routing, form validation, Fluent UI components, API consumption
API Gateway / Host	ASP.NET Core 8 Web API	HTTP endpoints, JWT auth middleware, CORS, Swagger/OpenAPI
Application	MediatR (CQRS), FluentValidation, AutoMapper	Commands, queries, handlers, DTOs, validation, mapping
Domain	C# class library (entities, enums, domain events, interfaces)	Business rules, domain models, repository interfaces
Infrastructure	EF Core, SQL Server, Hangfire, Serilog, Azure Blob	Data access, migrations, background jobs, logging, file storage

15.2 Key Patterns & Rationale

- Clean Architecture: dependency inversion ensures domain logic is independent of infrastructure; supports testability and maintainability.
- Vertical Slice Architecture: features organized by use case (e.g., Leave/ApplyLeave, Leave/ApproveLeave) rather than technical layer; reduces cross-cutting coupling.
- CQRS via MediatR: separates read and write operations; each handler is a single-responsibility unit.
- Repository + Unit of Work: abstracts data access; EF Core DbContext implements Unit of Work.
- Soft Deletes: global query filters on IsDeleted; preserves data integrity and audit trail.
- Audit Interceptor: EF Core SaveChanges interceptor auto-populates audit columns and writes to AuditLog.
- Background Jobs (Hangfire): scheduled leave accrual, approval escalation, payroll processing, email dispatch.
- Structured Logging (Serilog): JSON logs to Azure Application Insights for monitoring and alerting.

15.3 Frontend Architecture

- Angular standalone components with lazy-loaded feature modules per domain (employee, leave, payroll, etc.).
- Fluent UI Blazor-inspired design via @fluentui/web-components or custom Fluent theme on Angular Material.
- NgRx or Angular Signals for state management in complex modules (payroll, leave).
- Route guards for RBAC enforcement; HTTP interceptor for JWT attachment and refresh.
- Reactive forms with client-side validation matching FluentValidation rules.

15.4 Azure Hosting Topology

Azure Service	Purpose
Azure App Service (API)	Host ASP.NET Core Web API
Azure App Service (Web)	Host Angular SPA (or Azure Static Web Apps)
Azure SQL Database	Primary relational data store (Business Critical tier for production)
Azure Blob Storage	Document and payslip PDF storage
Azure Application Insights	APM, logging, alerting
Azure Key Vault	Secrets management (connection strings, JWT keys, encryption keys)
Azure Cache for Redis (optional)	Session/token caching at scale

16. Assumptions, Constraints & Dependencies

ID	Type	Description
A-01	Assumption	Client will provide Nigerian public holiday calendar annually.
A-02	Assumption	Client will provide current PAYE tax

		bands and update when Finance Act changes.
A-03	Assumption	Client has existing employee data available for migration in CSV format.
A-04	Assumption	Employees have corporate email addresses for authentication.
A-05	Assumption	Client IT team will provision Azure resources per architecture spec.
A-06	Assumption	Biometric device vendor will provide API/SDK for clock event integration.
A-07	Constraint	Nigeria-only payroll compliance in scope; no multi-currency payroll.
A-08	Constraint	Web-responsive only; no native mobile apps in Phases 1–3.
A-09	Constraint	Single-tenant deployment per client instance (not multi-tenant SaaS in v1).
A-10	Constraint	English (en-NG) only UI in initial releases.
A-11	Dependency	Azure subscription with appropriate service quotas.
A-12	Dependency	SMTP relay or Azure Communication Services for email delivery.
A-13	Dependency	SSL certificate for custom domain.
A-14	Dependency	Client HR team availability for UAT and policy decisions during each phase.
A-15	Dependency	Client Finance team for payroll validation and bank file format confirmation.

17. Release Plan / Phased Roadmap

Phase	Milestone	Key Deliverables	Exit Criteria
Phase 1	M1: Foundation	Project scaffold, DB schema, auth, RBAC, CI/CD pipeline	Users can authenticate; roles assigned; audit logging operational
Phase 1	M2: Employee Core	Employee CRUD, departments, positions, org chart, documents	HR can onboard employees; directory searchable; profiles viewable
Phase 1	M3: Leave & ESS/MSS	Leave types/policies, apply/approve, balances, dashboards, notifications	End-to-end leave workflow live; 95% of employees onboarded
Phase 1	M4: UAT & Go-Live	UAT, bug fixes, data migration, training, production deployment	Phase 1 sign-off; go-live approval from HR & IT
Phase 2	M5: Attendance	Clock-in/out, shifts, timesheets, device API, attendance reports	Attendance captured for 90% of employees
Phase 2	M6: Payroll	Salary structures, PAYE/Pension/NHF/NSITF,	First live payroll run completed with < 2%

		payroll run, payslips, bank file	adjustments
Phase 2	M7: Analytics	Dashboards, headcount, attrition, leave/payroll reports	Executive dashboard live; reports validated by Finance
Phase 2	M8: UAT & Go-Live	UAT, parallel payroll run, go-live	Phase 2 sign-off; parallel payroll reconciled
Phase 3	M9: Recruitment	Job postings, ATS pipeline, offer letters, onboarding checklists	First hire processed end-to-end in system
Phase 3	M10: Performance	Review cycles, goals, appraisals, 360 feedback	First review cycle completed in system
Phase 3	M11: UAT & Go-Live	UAT, training, go-live	Phase 3 sign-off
Later	M12+	Training & Development, Benefits & Compliance, SSO, mobile apps	Per separate PRD

18. Risks & Mitigations

Risk ID	Risk	Likelihood	Impact	Mitigation
RK-01	Incorrect PAYE/tax calculation due to regulatory changes	Medium	High	Configurable tax bands; annual review process with Finance; automated unit tests against known scenarios
RK-02	Low employee adoption of self-service	Medium	High	Change management program; simple UX; training sessions; manager champions
RK-03	Data migration errors from legacy spreadsheets	High	High	Migration validation scripts; parallel run period; HR sign-off on migrated data
RK-04	Scope creep across phases	High	Medium	Strict MoSCoW prioritization; change request process; phase gate reviews
RK-05	Biometric device integration incompatibility	Medium	Medium	Vendor-agnostic API; early POC with client's device; manual entry fallback
RK-06	Performance degradation at scale	Low	High	Load testing at 2x expected headcount; Azure SQL tuning; pagination on all lists
RK-07	NDPR non-compliance	Low	High	DPO review; encryption; access controls; privacy impact assessment

RK-08	Payroll run failure mid-processing	Low	High	Transactional payroll processing; idempotent runs; rollback capability; Hangfire retry policies
RK-09	Key person dependency on client HR team for UAT	Medium	Medium	Early engagement; documented test cases; dedicated UAT window with backups
RK-10	Azure service outage	Low	High	Azure SLA reliance; DR plan with geo-redundant backups; communication plan
RK-11	Security breach / unauthorized access	Low	Critical	Penetration testing; OWASP compliance; MFA readiness; audit logging; incident response plan
RK-12	Bank file format rejection	Medium	High	Early confirmation of bank template with Finance; test file submission before go-live

19. Glossary of Terms

Term	Definition
CRAS	Consolidated Relief Allowance — deducted from taxable income before PAYE calculation per Nigeria Tax Act.
ESS	Employee Self-Service — portal for employees to manage their own HR tasks.
FIRS	Federal Inland Revenue Service — Nigeria's federal tax authority.
MSS	Manager Self-Service — portal for line managers to approve requests and view team data.
NDPR	Nigeria Data Protection Regulation — data privacy law governing personal data processing.
NHF	National Housing Fund — 2.5% employee contribution on basic salary for housing finance.
NIN	National Identification Number — Nigeria's national ID number.
NIBSS	Nigeria Inter-Bank Settlement System — facilitates electronic fund transfers between banks.
NSITF	Nigeria Social Insurance Trust Fund — employer contribution for employee compensation insurance.
PAYE	Pay As You Earn — progressive income tax deducted at source from employee salaries.
PFA	Pension Fund Administrator — licensed entity managing employee pension contributions.
PENCOM	National Pension Commission — regulatory body for pension in Nigeria.

PFI	Pension Fund Custodian — holds pension assets on behalf of PFAs.
RBAC	Role-Based Access Control — authorization model based on user roles.
RC Number	Registration Certificate Number — company registration identifier from CAC.
SLA	Service Level Agreement — agreed response/resolution time for workflow actions.
SSO	Single Sign-On — authentication via external identity provider (e.g., Azure AD).
TIN	Tax Identification Number — issued by FIRS for tax purposes.
WAT	West Africa Time — UTC+1, Nigeria's standard timezone.

20. Appendix

Appendix A: RBAC Permission Matrix

Permission / Module	Employee	Line Manager	HR Admin	HR Manager	Payroll Officer	System Admin	Executive
View own profile	✓	✓	✓	✓	✓	—	—
Edit own profile (limited)	✓	✓	✓	✓	✓	—	—
View team profiles	—	✓	✓	✓	—	—	—
View all employee profiles	—	—	✓	✓	✓	—	—
Create/edit employee records	—	—	✓	✓	—	—	—
Deactivate employee	—	—	✓	✓	—	—	—
Manage departments/positions	—	—	✓	✓	—	—	—
Upload/view own documents	✓	✓	✓	✓	✓	—	—
Upload/view all documents	—	—	✓	✓	—	—	—
Apply for leave	✓	✓	✓	✓	✓	—	—
Approve team leave	—	✓	✓	✓	—	—	—
Approve all leave / override	—	—	✓	✓	—	—	—
Configure leave types/policies	—	—	✓	✓	—	—	—
Adjust leave balances	—	—	✓	✓	—	—	—
Clock in/out	✓	✓	✓	✓	✓	—	—
View team attendance	—	✓	✓	✓	—	—	—
Manage attendance/shifts	—	—	✓	✓	—	—	—
View own payslips	—	—	—	—	✓	—	—
Manage salary structures	—	—	—	—	✓	—	—
Run/approve payroll	—	—	—	✓	✓	—	—
Export bank payment file	—	—	—	—	✓	—	—

Manage recruitment	—	—	✓	✓	—	—	—
Manage performance reviews	—	✓	✓	✓	—	—	—
View HR reports	—	—	✓	✓	✓	—	—
View executive dashboards	—	—	—	✓	—	—	✓
Manage roles/permissions	—	—	—	—	—	✓	—
View audit logs	—	—	—	—	—	✓	—
System configuration	—	—	—	—	—	✓	—

Appendix B: Nigeria PAYE Tax Bands (2024 Finance Act Reference)

Note: Tax bands must be configurable in the system to accommodate annual Finance Act changes. Below reflects the 2024 reference structure.

Annual Taxable Income (₦)	Tax Rate
First ₦800,000	0%
Next ₦2,200,000	15%
Next ₦9,000,000	18%
Next ₦13,000,000	21%
Next ₦25,000,000	23%
Above ₦50,000,000	25%

Consolidated Relief Allowance (CRA): Higher of ₦200,000 or 1% of gross income, plus 20% of gross income. Applied before band calculation.

Appendix C: Nigeria Statutory Deductions Summary

Deduction	Rate	Paid By	Basis	Regulatory Body
PAYE	Progressive (see Appendix B)	Employee	Taxable income after CRA	FIRS / State IRS
Pension	8%	Employee	Monthly emoluments (basic + housing + transport)	PENCOM
Pension	10%	Employer	Monthly emoluments	PENCOM
NHF	2.5%	Employee	Basic salary	Federal Mortgage Bank
NSITF	1% (configurable)	Employer	Total monthly emoluments	NSITF

Appendix D: Sample Report Output Fields

Report	Sample Columns
Payroll Register	Employee ID, Name, Department, Basic, Housing, Transport, Gross, PAYE, Pension (EE), Pension (ER), NHF, NSITF, Other Deductions, Net Pay
Headcount	Department, Location, Active Count, On Leave, New Hires (MTD), Exits (MTD)
Attrition	Period, Department, Opening HC, Exits, Attrition Rate,

	Voluntary, Involuntary
Leave Utilization	Employee, Department, Leave Type, Entitled, Used, Remaining, Utilization %
Bank Payment File	S/N, Employee Name, Bank Code, Account Number, Amount (₦), Narration

Appendix E: API Endpoint Conventions

Convention	Standard
Base URL	https://api.employee360.{client-domain}/api/v1
Authentication	Bearer JWT in Authorization header
Versioning	URL path versioning (/api/v1/...)
Pagination	Query params: page, pageSize (default 20, max 100)
Sorting	Query param: sortBy, sortDirection (asc/desc)
Error format	RFC 7807 Problem Details (application/problem+json)
Date format	ISO 8601 (YYYY-MM-DDTHH:mm:ss+01:00)
Currency	Decimal (18,2) stored; displayed as ₦ with 2 decimal places

Appendix F: API Module Endpoints (Reference)

Module	Method	Endpoint	Description	Phase
Auth	POST	/api/v1/auth/login	Authenticate user; return JWT	Phase 1
Auth	POST	/api/v1/auth/refresh	Refresh access token	Phase 1
Auth	POST	/api/v1/auth/forgot-password	Initiate password reset	Phase 1
Employees	GET	/api/v1/employees	List employees (paginated, filterable)	Phase 1
Employees	POST	/api/v1/employees	Create employee	Phase 1
Employees	GET	/api/v1/employees/{id}	Get employee by ID	Phase 1
Employees	PUT	/api/v1/employees/{id}	Update employee	Phase 1
Employees	GET	/api/v1/employees/org-chart	Get org chart tree	Phase 1
Departments	CRUD	/api/v1/departments	Manage departments	Phase 1
Positions	CRUD	/api/v1/positions	Manage positions	Phase 1
Documents	POST	/api/v1/employees/{id}/documents	Upload document	Phase 1
Leave	POST	/api/v1/leave/requests	Submit leave request	Phase 1
Leave	PUT	/api/v1/leave/requests/{id}/approve	Approve leave	Phase 1
Leave	GET	/api/v1/leave/balances/{employeeId}	Get leave balances	Phase 1
Leave	GET	/api/v1/leave/calendar	Team leave calendar	Phase 1
Attendance	POST	/api/v1/attendance/clock-in	Clock in	Phase 2
Attendance	POST	/api/v1/attendance/clock-out	Clock out	Phase 2
Attendance	POST	/api/v1/attendance/device-events	Biometric device ingestion	Phase 2
Payroll	POST	/api/v1/payroll/runs	Initiate payroll	Phase 2

			run	
Payroll	GET	/api/v1/payroll/runs/{id}/payslips	List payslips for run	Phase 2
Payroll	GET	/api/v1/payroll/payslips/{id}/download	Download payslip PDF	Phase 2
Payroll	GET	/api/v1/payroll/runs/{id}/bank-file	Export bank payment file	Phase 2
Reports	GET	/api/v1/reports/{reportId}	Generate report	Phase 2
Recruitment	CRUD	/api/v1/jobs	Manage job postings	Phase 3
Recruitment	CRUD	/api/v1/candidates	Manage candidates	Phase 3
Performance	CRUD	/api/v1/reviews	Manage performance reviews	Phase 3
Admin	GET	/api/v1/audit-logs	Query audit logs	Phase 1
Admin	CRUD	/api/v1/roles	Manage roles and permissions	Phase 1

Appendix G: UI Screen Inventory

Screen ID	Screen Name	Module	Roles	Phase
SCR-001	Login	Auth	All	Phase 1
SCR-002	Forgot Password	Auth	All	Phase 1
SCR-003	ESS Dashboard	Self-Service	Employee	Phase 1
SCR-004	MSS Dashboard	Self-Service	Line Manager	Phase 1
SCR-005	Employee Directory	Employee	HR Admin, Manager	Phase 1
SCR-006	Employee Profile	Employee	All (scoped)	Phase 1
SCR-007	Create/Edit Employee	Employee	HR Admin	Phase 1
SCR-008	Org Chart	Employee	All	Phase 1
SCR-009	Department Management	Employee	HR Admin	Phase 1
SCR-010	Position Management	Employee	HR Admin	Phase 1
SCR-011	Apply for Leave	Leave	Employee	Phase 1
SCR-012	My Leave Requests	Leave	Employee	Phase 1
SCR-013	Leave Approval Queue	Leave	Manager, HR	Phase 1
SCR-014	Team Leave Calendar	Leave	Manager	Phase 1
SCR-015	Leave Policy Configuration	Leave	HR Admin	Phase 1
SCR-016	Leave Balance Management	Leave	HR Admin	Phase 1
SCR-017	Public Holiday Calendar	Admin	HR Admin	Phase 1
SCR-018	Company Settings	Admin	HR Admin, System Admin	Phase 1
SCR-019	Role & Permission Management	Admin	System Admin	Phase 1
SCR-020	Audit Log Viewer	Admin	System Admin, HR Manager	Phase 1
SCR-021	Clock In/Out	Attendance	Employee	Phase 2
SCR-022	Attendance Log	Attendance	Employee, Manager, HR	Phase 2

SCR-023	Shift Management	Attendance	HR Admin	Phase 2
SCR-024	Timesheet Entry	Attendance	Employee	Phase 2
SCR-025	Salary Structure Management	Payroll	Payroll Officer	Phase 2
SCR-026	Employee Salary Assignment	Payroll	Payroll Officer	Phase 2
SCR-027	Payroll Run	Payroll	Payroll Officer	Phase 2
SCR-028	Payroll Approval	Payroll	HR Manager, Finance	Phase 2
SCR-029	My Payslips	Payroll	Employee	Phase 2
SCR-030	Reports Dashboard	Reports	HR, Executive	Phase 2
SCR-031	Job Postings	Recruitment	HR Admin	Phase 3
SCR-032	Candidate Pipeline	Recruitment	HR Admin	Phase 3
SCR-033	Onboarding Checklist	Recruitment	HR Admin, Employee	Phase 3
SCR-034	Review Cycle Management	Performance	HR Admin	Phase 3
SCR-035	My Goals & Appraisal	Performance	Employee	Phase 3
SCR-036	Team Appraisals	Performance	Line Manager	Phase 3

Appendix H: Database Indexing Strategy

Table	Index	Rationale
Employee	IX_Employee_DepartmentId_Status	Directory filtering by department and status
Employee	IX_Employee_ManagerId	Manager team queries and org chart
Employee	IX_Employee_Email (unique)	Login lookup
LeaveRequest	IX_LeaveRequest_EmployeeId_Status	Employee leave history and pending queries
LeaveRequest	IX_LeaveRequest_ApproverId_Status	Manager approval queue
AttendanceRecord	IX_Attendance_EmployeeId_Date (unique)	Daily attendance lookup
Payslip	IX_Payslip_EmployeeId_Period	Payslip history retrieval
AuditLog	IX_AuditLog_EntityName_EntityId	Entity change history
AuditLog	IX_AuditLog_Timestamp	Date-range audit queries

Appendix I: Email Notification Templates

Template ID	Trigger	Recipient	Subject Line
EML-001	Leave submitted	Line Manager	Leave Request from {EmployeeName} — Action Required
EML-002	Leave approved	Employee	Your Leave Request has been Approved
EML-003	Leave rejected	Employee	Your Leave Request has been Declined
EML-004	Leave escalated	Escalation Manager	Escalated: Pending Leave Approval for {EmployeeName}
EML-005	Welcome / account created	New Employee	Welcome to Employee360 — Set Up Your Account
EML-006	Password reset	User	Employee360 Password Reset Request

EML-007	Payslip available	Employee	Your Payslip for {Period} is Ready
EML-008	Payroll pending approval	HR Manager	Payroll Run {Period} — Approval Required
EML-009	Profile change approved	Employee	Your Profile Update has been Approved
EML-010	Onboarding task assigned	Task Assignee	Onboarding Task Assigned: {TaskName}